

Expanders Give a Negative Answer to the Finite-Complexity-Rate Inductive-Limit Question

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Abstract

Tian and Yu asked whether every countable discrete group is an inductive limit of groups with finite complexity rates. We first verify the supplied direct-union reduction: under injective bonding maps, the question is equivalent to asking whether every finitely generated group has finite complexity rate. We then give a counterexample.

The key observation is quantitative. If a finitely generated group coarsely contains a bounded-degree expander sequence, then every family of maps used in the definition of complexity rate has unit-scale variation at least a constant multiple of $1/p$ in the target ℓ^{2p} . This follows from the L_q Poincaré inequality for expanders with constant $O(q)$. Hence such a group has decay rate exactly 1, or equivalently infinite complexity rate. Osajda constructed finitely presented groups—indeed fundamental groups of closed aspherical manifolds—that coarsely contain expanders. One of these groups is therefore a finitely presented counterexample.

Finally, finite presentability shows that the same example cannot be a filtered colimit of finite-complexity-rate groups even when arbitrary, non-injective bonding homomorphisms are allowed. Thus the answer to the question is negative under both standard interpretations of “inductive limit.”

1 The question and the rate condition

Tian and Yu prove the strong Novikov conjecture for countable groups with finite complexity rate and ask:

“It is natural to ask whether every countable discrete group is an inductive limit of groups with finite complexity rates.”

Their rate condition is Definition 1.4 of [4]; Definition 1.3 there only defines the iterated logarithms.

For the present argument, only the following consequence of the definition is needed. If $\text{CR}(\Gamma) < \infty$, then there are a finite tuple

$$\mathbf{a} = (a_0, a_1, \dots, a_k) > 1, \quad a_0 \geq 1, \quad a_j \geq 0,$$

in the lexicographic order used in [4], a constant $C > 0$, numbers $N_p > 0$, and maps

$$\phi_p : \Gamma \longrightarrow S(\ell^{2p}(\mathbb{N})) \quad (p \in \mathbb{N})$$

such that

$$d(x, y) \leq 1 \implies \|\phi_p(x) - \phi_p(y)\|_{2p} \leq \varepsilon_p, \tag{1.1}$$

$$d(x, y) \geq N_p \implies \|\phi_p(x) - \phi_p(y)\|_{2p} \geq 1, \tag{1.2}$$

where

$$\varepsilon_p = \frac{C}{p^{a_0}(\log p)^{a_1} \cdots (\log_k p)^{a_k}}.$$

The truncated convention for the iterated logarithms at small values of p is irrelevant here. Since $\mathbf{a} > 1$ lexicographically,

$$p\varepsilon_p \longrightarrow 0. \quad (1.3)$$

Indeed, either $a_0 > 1$, or $a_0 = 1$ and the first nonzero logarithmic exponent is positive.

The definition is independent of the chosen left-invariant proper metric on Γ [4]. For finitely generated groups we therefore use a word metric.

2 Audit of the supplied partial result

Lemma 2.1 (Subgroup heredity). *If a countable group Γ has finite complexity rate, then every subgroup $\Lambda \leq \Gamma$ has finite complexity rate.*

Proof. Restrict a left-invariant proper metric on Γ to Λ . It remains left-invariant and proper. Restricting each witness ϕ_p preserves both the variation estimate and the large-scale separation estimate. Metric independence of the definition then gives the intrinsic assertion for Λ . $\square$

Proposition 2.2 (Direct-union reduction). *For a countable group Γ , the following are equivalent.*

1. Γ is a directed union of subgroups with finite complexity rate.
2. Every finitely generated subgroup of Γ has finite complexity rate.

Proof. Suppose $\Gamma = \bigcup_{i \in I} \Gamma_i$, with I directed and all Γ_i of finite complexity rate. A finite generating set of a subgroup $H \leq \Gamma$ is contained in one stage, so H is a subgroup of that stage and Lemma 2.1 applies.

Conversely, the finitely generated subgroups of Γ , ordered by inclusion, form a directed set: H_1 and H_2 are contained in $\langle H_1 \cup H_2 \rangle$. Their union is Γ . $\square$

Thus the main reduction in the supplied packet is correct. Its additional observation about non-injective systems is also correct: a filtered colimit of torsion-free groups is torsion-free. If the class of an element g_i has finite order in the colimit, then a later image of g_i has finite order; torsion-freeness of that later stage forces the image, and hence the colimit class, to be trivial. In particular, torsion groups cannot be obtained as filtered colimits of free groups.

3 The expander obstruction

We use spectral expanders. Let $X = (V, E)$ be a finite connected d -regular graph, and let A_X be the normalized adjacency operator. We say that a sequence (X_n) is a spectral expander sequence with gap $\sigma > 0$ if $|V(X_n)| \rightarrow \infty$ and

$$1 - \lambda_2(A_{X_n}) \geq \sigma$$

for every n .

We say that (X_n) *coarsely embeds into* a finitely generated group Γ if there are maps $\iota_n : V(X_n) \rightarrow \Gamma$ and nondecreasing functions $\rho_-, \rho_+ : [0, \infty) \rightarrow [0, \infty)$, independent of n , such that $\rho_-(t) \rightarrow \infty$ and

$$\rho_-(d_{X_n}(u, v)) \leq d_\Gamma(\iota_n(u), \iota_n(v)) \leq \rho_+(d_{X_n}(u, v)). \quad (3.1)$$

The analytic input is the following form of Matoušek's extrapolated Poincaré inequality. It is Corollary 4.5 of Naor–Silberman [2]; see also Matoušek [1].

Lemma 3.1 (L_q Poincaré inequality). *Let $X = (V, E)$ be a finite d -regular graph whose normalized adjacency operator has spectral gap at least $\sigma > 0$. For every $q \geq 2$ and every map $f : V \rightarrow \ell^q$,*

$$\left(\frac{1}{|V|^2} \sum_{u,v \in V} \|f(u) - f(v)\|_q^q \right)^{1/q} \leq \frac{2q}{\sqrt{\sigma}} \left(\frac{1}{d|V|} \sum_{u \in V} \sum_{v \sim u} \|f(u) - f(v)\|_q^q \right)^{1/q}. \quad (3.2)$$

Theorem 3.2 (Expanders force the critical $1/p$ rate). *Let Γ be finitely generated. Suppose a d -regular spectral expander sequence with gap $\sigma > 0$ coarsely embeds into Γ . Let*

$$\phi_p : \Gamma \rightarrow S(\ell^{2p}(\mathbb{N}))$$

be any maps satisfying (1.1) and (1.2), with unit-scale variations ε_p . Then there is a constant $c > 0$, independent of p , such that

$$p\varepsilon_p \geq c \quad \text{for every } p \in \mathbb{N}. \quad (3.3)$$

Consequently, $\text{CR}(\Gamma) = \infty$ (equivalently, $\text{DR}(\Gamma) = 1$ in the notation of [4]).

Proof. Let

$$L = \max\{1, \lceil \rho_+(1) \rceil\}.$$

Because the word metric on Γ is geodesic, (1.1) implies

$$d_\Gamma(x, y) \leq L \implies \|\phi_p(x) - \phi_p(y)\|_{2p} \leq L\varepsilon_p. \quad (3.4)$$

Fix p and put $q = 2p$. Choose an integer $R_p \geq 1$ such that $\rho_-(R_p) \geq N_p$. Since the degrees of the graphs are bounded, the cardinality of every radius- $(R_p - 1)$ ball is at most

$$B_p = 1 + d \sum_{j=0}^{R_p-2} (d-1)^j,$$

with the empty sum interpreted as zero. Choose n so large that $|V(X_n)| \geq 2B_p$. Then at least half of the ordered pairs $(u, v) \in V(X_n)^2$ satisfy $d_{X_n}(u, v) \geq R_p$.

Set $f = \phi_p \circ \iota_n$. For every such far pair, (3.1) and (1.2) give

$$\|f(u) - f(v)\|_q \geq 1.$$

Therefore

$$\left(\frac{1}{|V(X_n)|^2} \sum_{u,v \in V(X_n)} \|f(u) - f(v)\|_q^q \right)^{1/q} \geq 2^{-1/q}. \quad (3.5)$$

If $u \sim v$, then (3.1) gives $d_\Gamma(\iota_n(u), \iota_n(v)) \leq \rho_+(1) \leq L$, so (3.4) yields

$$\|f(u) - f(v)\|_q \leq L\varepsilon_p.$$

Applying Lemma 3.1 and (3.5),

$$2^{-1/q} \leq \frac{2q}{\sqrt{\sigma}} L\varepsilon_p.$$

Since $q = 2p$ and $2^{-1/(2p)} \geq 2^{-1/2}$,

$$p\varepsilon_p \geq \frac{\sqrt{\sigma}}{4L} 2^{-1/(2p)} \geq \frac{\sqrt{\sigma}}{4\sqrt{2}L}.$$

This proves (3.3).

If $\text{CR}(\Gamma) < \infty$, the defining variations satisfy (1.3), contradicting (3.3). Hence $\text{CR}(\Gamma) = \infty$. Tian and Yu show that every countable group has decay rate at least 1, so the decay rate here is exactly 1. $\square$

4 A finitely presented counterexample and arbitrary filtered colimits

Osajda constructed finitely generated groups whose Cayley graphs contain isometrically embedded expander sequences [3, Theorem 3.2 and Corollary 3.3]. His construction applies to fixed-degree regular expander families; for such families, combinatorial expansion and Cheeger's inequality give the uniform normalized spectral gap required in Theorem 3.2. More strongly, he obtained the following consequence of Sapir's asphericity-preserving Higman embedding construction:

There exist closed aspherical manifolds of dimension at least four whose fundamental groups contain coarsely embedded expanders [3, Corollary 3.5].

A fundamental group of a closed manifold is finitely presented. Combining this with Theorem 3.2 gives:

Corollary 4.1. *There exists a finitely presented group G with $\text{CR}(G) = \infty$.*

For injective inductive limits, Proposition 2.2 already makes G a counterexample. Finite presentability also handles arbitrary bonding homomorphisms.

Lemma 4.2 (A finitely presented colimit is a retract of a stage). *Let (G_i, α_{ij}) be a directed system of groups and let*

$$G \cong \varinjlim_i G_i.$$

If G is finitely presented, then for some i there is a homomorphism $s : G \rightarrow G_i$ such that the canonical map $\lambda_i : G_i \rightarrow G$ satisfies $\lambda_i \circ s = \text{id}_G$. In particular, G is isomorphic to a subgroup of G_i .

Proof. Write

$$G = \langle x_1, \dots, x_m \mid r_1, \dots, r_k \rangle.$$

Choose representatives of the finitely many generators in stages of the system, and then pass to one common later stage G_i . Each word r_j in those representatives maps to the identity in the colimit. Equality in a filtered colimit is witnessed at a later stage, so after passing to a common later stage $G_{i'}$, all finitely many relators are simultaneously trivial. Sending x_l to its chosen representative in $G_{i'}$ therefore defines a homomorphism $s : G \rightarrow G_{i'}$. The composite $\lambda_{i'} \circ s$ fixes every generator x_l , hence is the identity. Thus s is injective. $\square$

Theorem 4.3 (Negative answer under both interpretations). *The question of Tian and Yu has a negative answer.*

1. *There is a countable, finitely presented group that is not a directed union of groups with finite complexity rates.*

2. *The same group is not a filtered colimit of groups with finite complexity rates, even if the bonding homomorphisms are arbitrary and need not be injective.*

Proof. Let G be the finitely presented group from Corollary 4.1.

For the first assertion, if G were a directed union of finite-complexity-rate subgroups, its finite generating set would lie in one stage. That stage would contain all of G , so G itself would have finite complexity rate, a contradiction.

For the second assertion, suppose $G \cong \varinjlim_i G_i$ with every G_i of finite complexity rate. By Lemma 4.2, G embeds in one G_i . Subgroup heredity, Lemma 2.1, would then imply $\text{CR}(G) < \infty$, again a contradiction. $\square$

5 Literature status

The exact notion of complexity rate was introduced only in May 2026. Searches by the paper title, arXiv identifier, the phrase “finite complexity rate,” and the inductive-limit question did not locate a later paper or note explicitly answering the question as of June 21, 2026. The counterexample above is a synthesis of two older ingredients: the linear-in- q Poincaré bound for L_q targets and the existence of finitely presented groups coarsely containing expanders. No claim of priority beyond that synthesis is made here.

References

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- [4] Geng Tian and Guoliang Yu, Embedding complexity into Banach spaces and the strong Novikov conjecture, arXiv:2605.12930v1, 2026.